

WEEK 3

Advanced Cybersecurity Practical Report

Network Discovery • Nmap • Wireshark • Vulnerability Assessment • Security Hardening

Lab Environment

Kali Linux → Security Testing / Scanning Machine

Metasploitable 2 → Authorized Vulnerable Target

Prepared from the completed Week 3 assignment and practical observations.

1. Executive Summary

Week 3 focused on performing an end-to-end basic security assessment of an authorized laboratory network. The work covered network discovery, basic and advanced Nmap scanning, Wireshark traffic capture and analysis, Nmap-to-Wireshark investigation, vulnerability assessment, security hardening, and a final mini security assessment.

The lab used Kali Linux at 192.168.27.128 and Metasploitable 2 at 192.168.27.129 on the 192.168.27.0/24 network. Nmap identified numerous exposed services, including legacy and intentionally vulnerable services. Wireshark was used to examine DNS, TCP, UDP, ARP and related traffic. Selected services were then disabled and the before/after results showed reduced network exposure.

2. Lab Environment

Component	Role	IP Address	Network
Kali Linux	Security testing / scanning machine	192.168.27.128	192.168.27.0/24
Metasploitable 2	Authorized vulnerable target / lab server	192.168.27.129	192.168.27.0/24

Kali Linux subnet: 255.255.255.0 (/24); default gateway: 192.168.27.2. Metasploitable 2 subnet: 255.255.255.0; default gateway: 192.168.27.2.

3. Task 7 – Network Discovery & Basic Nmap Scanning

Objective

Understand how a security analyst discovers devices and exposed network services.

Practical Work

- Network configuration was identified using ip addr and ip route.
- Host discovery was performed with an Nmap -sn scan against the authorized lab network.
- A basic Nmap scan was then performed against the authorized Metasploitable 2 target.

Observation: the host discovery scan identified active devices, while the basic scan identified open ports and associated services.

4. Task 8 – Advanced Nmap Security Assessment

Service and Version Detection

Command: nmap -sV 192.168.27.129

Port	Protocol	Service	Version / Info	Risk in assignment
21	TCP	FTP	vsftpd 2.3.4	High
22	TCP	SSH	OpenSSH 4.7p1	Medium
23	TCP	Telnet	Linux telnetd	High
80	TCP	HTTP	Apache 2.2.8	High
1099	TCP	Java RMI	GNU Classpath rmiregistry	High
1524	TCP	Bindshell	Metasploitable root shell	Critical
2049	TCP	NFS	RPC #100003	High
3306	TCP	MySQL	5.0.51a-3ubuntu5	High
5432	TCP	PostgreSQL	8.3.0–8.3.7	High
5900	TCP	VNC	Protocol 3.3	High

OS Detection

Command: sudo nmap -O 192.168.27.129

Nmap identified the target as a general-purpose Linux system in the Linux 2.6.x family, with an estimated kernel range of 2.6.9 to 2.6.33. The target was one network hop away and exposed multiple TCP services.

TCP SYN Scan

Command: sudo nmap -sS 192.168.27.129

The TCP SYN scan identified 22 open TCP ports and 978 closed TCP ports. The results were consistent with service/version detection and showed a broad attack surface, as expected for the intentionally vulnerable Metasploitable environment.

Limited UDP Scan

Command: sudo nmap -sU --top-ports 20 192.168.27.129

UDP port 53 (DNS) and UDP port 137 (NetBIOS Name Service) were identified as open. Several other ports were reported as open|filtered, meaning Nmap could not definitively determine their state because UDP responses may be absent or filtered.

Vulnerability NSE Scan

Command: nmap --script vuln 192.168.27.129

The vulnerability scan identified significant issues, including the vsftpd 2.3.4 backdoor vulnerability (CVE-2011-2523) and an insecure cryptographic configuration involving the SMTP service.

5. Service-Level Security Analysis

Service	Purpose	Security Concern	Suggested Security Measure
FTP / vsftpd 2.3.4	File transfer	Outdated/vulnerable service; unnecessary exposure	Upgrade; disable anonymous access; disable if unnecessary
SSH	Secure remote administration	Legacy version / excessive exposure	Upgrade and use strong authentication
Telnet	Remote command-line access	Plaintext protocol can expose credentials/session data	Disable Telnet; use SSH
HTTP / Apache 2.2.8	Web server	Legacy web-server version may contain vulnerabilities	Upgrade and remove unnecessary applications
Bindshell 1524	Training-environment root shell	Extremely high-privilege remote access	Disable/remove and restrict with firewall
MySQL	Database service	Exposed/legacy database service	Upgrade and restrict access
PostgreSQL	Database service	Exposed/legacy database service	Upgrade and restrict access
VNC	Remote GUI access	Unauthorized graphical access if poorly protected	Firewall restriction and secure tunneling

6. Task 9 – Wireshark Network Traffic Capture

Filters Used

- dns
- tcp
- udp
- icmp
- arp

- tcp.flags.syn == 1
- tcp.flags.reset == 1
- dns.flags.response == 0

The capture was analyzed for DNS, TCP, UDP and ARP packets. A TCP SYN-ACK retransmission was observed. A single retransmission does not by itself indicate malicious activity; it can occur when an expected acknowledgment is not received in time.

7. Task 10 – Nmap + Wireshark Investigation

Nmap command: nmap -sS 192.168.81.129

Nmap Result	Wireshark Evidence	Interpretation
Port 80 reachable/open	SYN from 192.168.0.102 to 142.250.134.94:80, followed by SYN/ACK	SYN/ACK indicates TCP port 80 is reachable and accepting connections.
Port 443 reachable/open	SYN from 192.168.0.102 to 52.123.128.14:443, followed by SYN/ACK	SYN/ACK indicates the TCP connection request was accepted at the handshake stage.

8. Task 11 – Advanced Network Traffic Investigation

Top Communicating Hosts

- Most active source IP: 192.168.0.102
- Most active destination IP: 192.168.0.102
- Communication types: TCP and TLS

DNS Investigation

- Source IP: 192.168.0.102
- Destination IP: 192.168.0.1 (local DNS resolver/default gateway router)
- Protocol: DNS over UDP port 53
- Observed query types: A and HTTPS

TCP Investigation

The local host generated SYN packets from high-numbered source ports to TCP port 443 on external addresses, including addresses associated with Google and Cloudflare. RST packets were also observed; these can occur when a socket is closed, traffic is blocked, or a remote endpoint refuses a connection.

Potentially Unusual Domain Activity

Domain	Observation	Security Significance
dump.mewcdn.online	Unusual DNS query structure was observed in the capture	Warrants further investigation based on the capture context
nekostream.site	Identified in the capture as a media/streaming-related domain	Should be reviewed in context; the capture alone does not establish malicious activity

9. Task 12 – Vulnerability Assessment

Finding	Evidence	Risk	Potential Impact	Recommendation
vsftpd 2.3.4 outdated / vulnerable	Nmap -sV: FTP on 21/TCP	Critical	Known backdoor can potentially enable unauthorized remote access	Disable if unnecessary; remove/upgrade
Telnet exposed	23/TCP open	High	Plaintext credentials/session data may be intercepted	Disable Telnet; use SSH

Finding	Evidence	Risk	Potential Impact	Recommendation
Outdated Apache	Apache 2.2.8 on 80/TCP	High	Known vulnerabilities may permit compromise	Upgrade and patch
Metasploitable root shell exposed	1524/TCP	Critical	Potential extremely high-privilege remote access	Disable and restrict with firewall
Large number of services exposed	Nmap identified many open TCP ports	High	Increases attack surface	Disable unnecessary services; apply firewall/ACL restrictions

10. Task 13 – Security Hardening & Before/After Testing

Hardening Actions

- FTP was disabled using: `sudo service vsftpd stop`
- Telnet was disabled using: `sudo service telnetd stop`
- The final assessment also documented disabling rexecd as a hardening action.

Security Issue	Before	Action Taken	After / Result
FTP service exposed	21/tcp open	FTP service disabled	21/tcp closed; reduced FTP exposure
Telnet exposed	23/tcp open	Telnet service disabled	23/tcp closed; removed unnecessary remote-access service
rexecd exposed	512/tcp open	rexecd disabled	Reduced legacy remote-service exposure

Attack Surface Reduction: Yes. The assignment reports that the attack surface was reduced because the number of exposed services decreased after hardening, and the before/after Nmap comparison showed the selected services were no longer accessible on their respective ports.

11. Task 14 – Final Mini Security Assessment

Objective

Perform a basic security assessment of an authorized lab network using network discovery, Nmap scanning, Wireshark traffic analysis, vulnerability assessment, and security hardening.

Network Discovery

Device	Role	IP Address
Kali Linux	Security testing / scanning machine	192.168.27.128
Metasploitable 2	Target / lab server	192.168.27.129

Key Services Identified

Port	Service	Version / Information
21	FTP	vsftpd 2.3.4
22	SSH	OpenSSH
23	Telnet	Linux telnetd
25	SMTP	SMTP service
80	HTTP	Apache 2.2.8
1524	Bindshell	Metasploitable root shell
3306	MySQL	5.0.51a
5432	PostgreSQL	8.3.x
5900	VNC	Protocol 3.3

Traffic Analysis

Wireshark showed protocol layers including Ethernet, IPv4, TCP, TLS and Data. The investigation also covered DNS, TCP connection attempts, RST packets, and domain-resolution activity.

Overall Security Findings

- Multiple unnecessary/exposed services increased the attack surface.
- Telnet exposed an insecure remote-access protocol.
- FTP exposed an outdated service with a known security issue in the lab.
- Legacy services and software increased security exposure.
- Hardening reduced exposure by disabling selected services.

Final Recommendations

- Disable unnecessary services.
- Replace insecure protocols with secure alternatives.
- Apply regular security updates and patches.
- Implement firewall restrictions and access controls.
- Continuously monitor network traffic and exposed services.

12. Week 3 Learning Outcomes

By completing the Week 3 practical work, the following skills were practiced:

- Network discovery and identification of active hosts.
- Basic, service/version, OS, SYN, UDP and NSE-based Nmap scanning.
- Interpreting open ports and exposed services.
- Analyzing packet captures with Wireshark filters.
- Connecting Nmap results with packet-level TCP handshake evidence.
- Investigating DNS and TCP communication patterns.
- Identifying security findings and classifying their risk.
- Applying basic security hardening and validating changes with before/after scans.
- Communicating technical findings in a structured security assessment.

13. Conclusion

Week 3 provided practical experience in a complete security-assessment workflow. The lab progressed from network discovery to service enumeration, traffic analysis, vulnerability identification, and hardening. The results demonstrated how exposed services contribute to attack surface and how disabling unnecessary services can reduce exposure. All scanning and testing described in this report were performed against the authorized laboratory environment.

Appendix – Key Commands

```
ip addr
ip route
nmap -sn /24
nmap 192.168.27.129
nmap -sV 192.168.27.129
```

```
sudo nmap -O 192.168.27.129
sudo nmap -sS 192.168.27.129
sudo nmap -sU --top-ports 20 192.168.27.129
nmap --script vuln 192.168.27.129
nmap -sS 192.168.81.129
sudo service vsftpd stop
sudo service telnetd stop
```